The seismoelectric method

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Introduction

This paper follows the structure of the IFP Energies nouvelles e-book dedicated to electromagnetic methods in Geophysics. it is meant as a companion to the online course part devoted to the Seismoelectric method.

Back to the theory

The seismoelectric method is a geophysical imaging technique relying on electro-kinetic conversions between seismic and electromagnetic waves. When a seismic wave travels throughout a fluid-filled porous rock, it sets in motion the fluid in the pores with respect to the solid rock matrix: the relative motion of the ions in the free electrolyte with respect to those adsorbed along the pore walls triggers an electrical response. Therefore, by combining a seismic source with a string of dipole receivers, it becomes possible to measure the electric field associated with these seismoelectric conversions.

Other conversions between mechanical and electromagnetic energy exist, such as piezoelectricity and resistivity modulation. These phenomena were also applied to geophysical exploration: for instance, the piezoelectric effect was applied to quartz and pegmatite veins detection in the late 1960's (Volarovitch *et al.*, 1969). These other phenomena will not be further addressed in this chapter.

Last but not least, the seismoelectric method should not be mistaken with its electro-seismic counterpart, which relies on electro-osmotic phenomena, i.e. the fluid motion induced by an applied potential across a porous material (see for instance Thompson *et al.*, 2007).

The Streaming Potential Coefficient

Applying a pressure gradient ΔP [Pa] to a porous rock sample induces the fluid in the pore space to flow. A local convection current originates from the movement of some of the ions in the electrolyte; this current is counterbalanced by a conduction current, thus ensuring the electro-neutrality of the medium. This ohmic current creates a potential difference ΔV [V], called the streaming potential.

In the case of laboratory experiments, it is possible to measure both the applied ΔP and the measured ΔV (Figure 1).

The ratio between these quantities is called the streaming potential coefficient, or SPC, often noted C (Equation 1).

$$C = \frac{\Delta V}{\Delta P}$$

Equation 1: the streaming potential coefficient

Any phenomenon creating a relative motion between the liquid and solid phases should therefore create a streaming potential: gravity flow, fluid-filled facture growth within a rock and the propagation of a seismic wave, which makes the seismoelectric method possible.

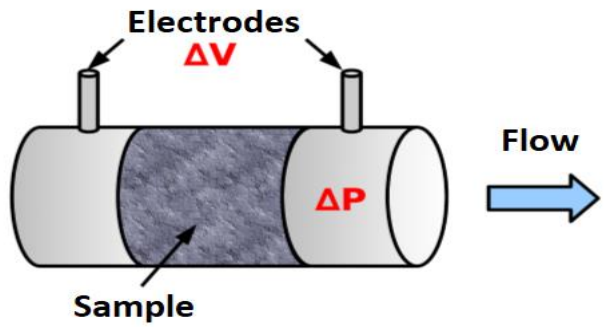


Figure 1: Experimental layout: the fluid flow induced by the pressure drop ΔP creates a potential difference ΔV measured by the electrode pair. From Warden 2012.

The Electrical Double Layer

Several models have been proposed to explain the mechanisms giving rise to the streaming coefficient. These models of Electrical Double Layer (EDL) or sometimes even Electrical Triple Layer (ETL) describe the way ions are organized in the pore fluid and along the mineral grains, at the microscopic scale.

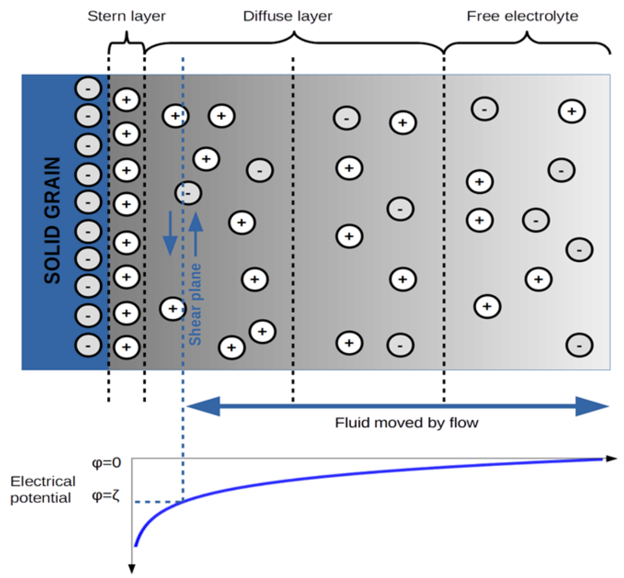


Figure 2: A simplified Electrical Double Layer (EDL) model.

Figure 2 shows such a model. The negatively charged mineral grain is depicted to the far left; to the right is the free electrolyte, where balance between positive and negative charges is achieved. Between these are a

series of intermediate layers.
The Stern layer contains positive ions (cations) adsorbed along the mineral grain. As the cations in the Stern layer do not compensate the negative surface charges, a diffuse layer exists, where cations outnumber the negative ions (anions). This excess of positive charges decreases exponentially until cations and anions are balanced in the free electrolyte. As a result, the electrical potential increases until it reaches 0 in the free electrolyte.

The different types of seismoelectric waves

Seismoelectric conversions mainly give rise to two types of waves:

- The **coseismic waves** (or type I waves).
- The **interface response** (or type II waves).

Let us consider a seismic source, such as a sledgehammer hitting a plate (see for instance Figure 3). When the subsurface contains fluid, the seismic wave gives rise to an accompanying electromagnetic wave that travels *within* the seismic disturbance. As it travels at the same velocity as the seismic wave that induces it, it is dubbed a ***coseismic* wave**.

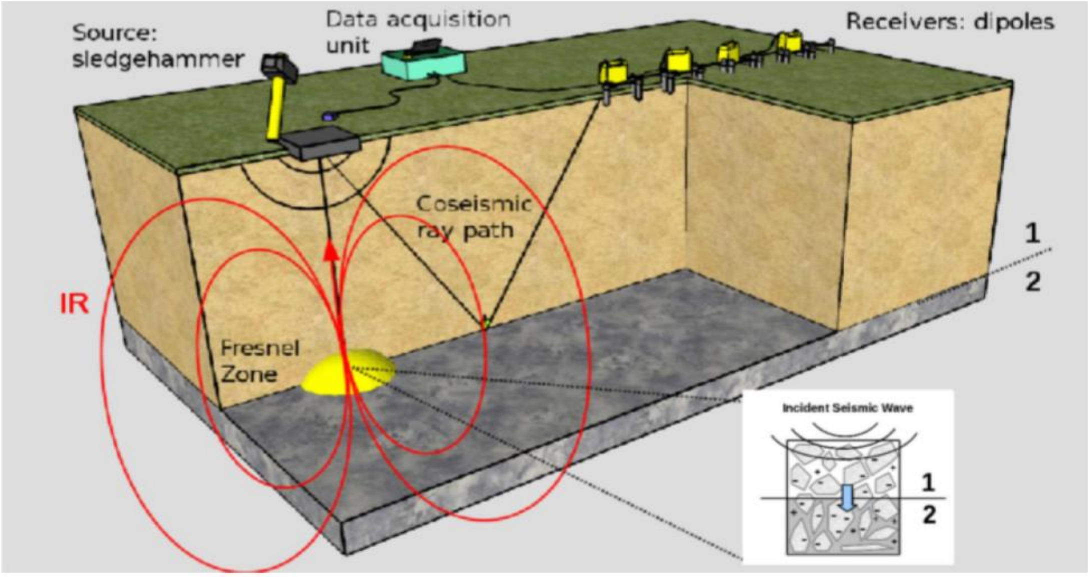


Figure 3: A simple seismoelectric data acquisition layout. The seismic source consists of a sledgehammer hitting a plate. The receivers are a string of dipoles connected to digitizers. A coseismic ray path is depicted in black in this Figure. When the incident seismic wave encounters the interface between layers 1 and 2, it gives rise to an interface response (IR) that can be modeled by a dipole oscillating at the first Fresnel zone: the IR is represented in red. From Warden et al., 2013.

Thus, taking a look at a seismic and a seismoelectric recording side by side (*e.g.* the synthetic recordings in Figure 4) reveals that some of the seismic arrivals show clear seismoelectric counterparts. In fact, seismoelectric dipole receivers were once foreseen as a possible alternative to conventional geophones.

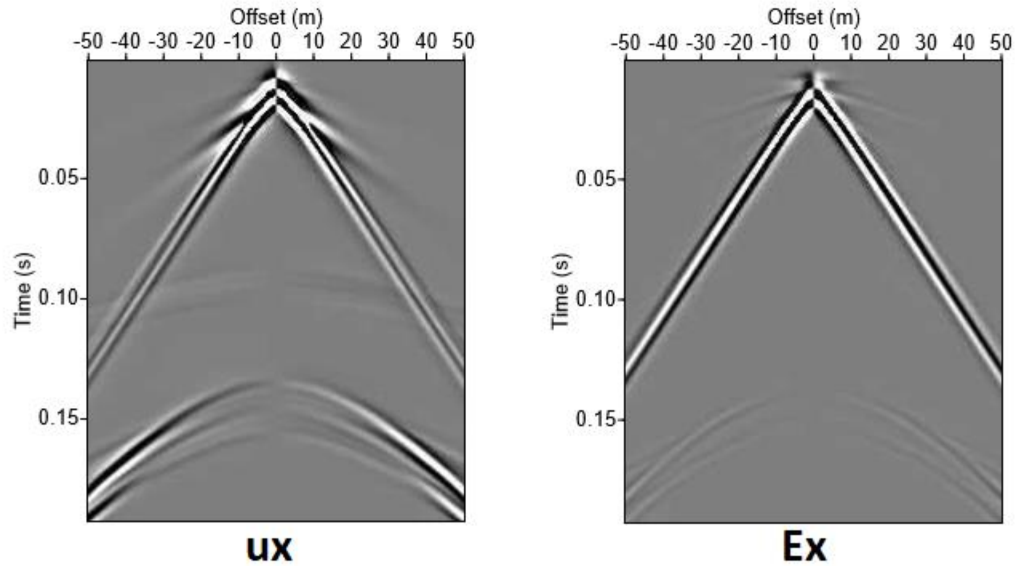


Figure 4: Synthetic seismic (to the left) and seismoelectric (to the right) recordings. These synthetic recordings were modeled for a simple two-layer tabular model consisting of a sand layer on top of a sandstone half-space. The source peak frequency is f_{peak}=120Hz.From Warden et al., 2012.

When the incident seismic wave encounters an interface between two media exhibiting different mechanical or electrical properties, it may give rise to an **interface response** (or IR), travelling with the velocity of an electromagnetic wave. As its velocity is much higher than seismic velocities, the IR looks like it seen almost instantaneously by all receivers; it therefore appears like a "horizontal" event in the seismoelectric recording.

An example of IR is given in Figure 5(c) for a synthetic recording. In this example, the amplitude of the IR has been artificially boosted; otherwise the IR would be concealed by the stronger coseismic arrivals seen in Figure 5(a) (the same recording as in Figure 4). In this example the IR can be approximated as a dipole oscillating at the first Fresnel zone at the interface: this means that its amplitude is not maximal right below the shot point, but instead reaches its maximum at a certain offset away from the source. The amplitude variations of the IR with respect to the offset are displayed in Figure 5(d). It can also be noted that the IR exhibits opposite polarities on either side of the shot point: this characteristic may be used to discriminate the IR from other arrivals.

Interface responses may provide valuable information about the subsurface at depth. However, their very weak amplitudes make them difficult to recover and analyze.
In the simple noise-free synthetic example detailed above (Figure 5(a)), the IR was concealed by the coseismic waves. One can imagine that it gets even trickier to recover such responses on noisy data acquired in the field. Section **Field Equipment and Data Processing** deals with the processing steps usually applied to extract IRs from seismoelectric recordings.

The seismoelectric theory of Pride (1994)

In this section, we briefly introduce Pride's theory (1994) for the coupled propagation of seismic and electromagnetic waves. One of Pride's key contribution was to consider the full set of Maxwell electromagnetic equations in order to derive the equations describing seismo-electromagnetic coupling. 50 years after the seminal yet incomplete work of Frenkel (1944), Pride's work rekindled the interest of the geophysical community for seismo-electric methods.

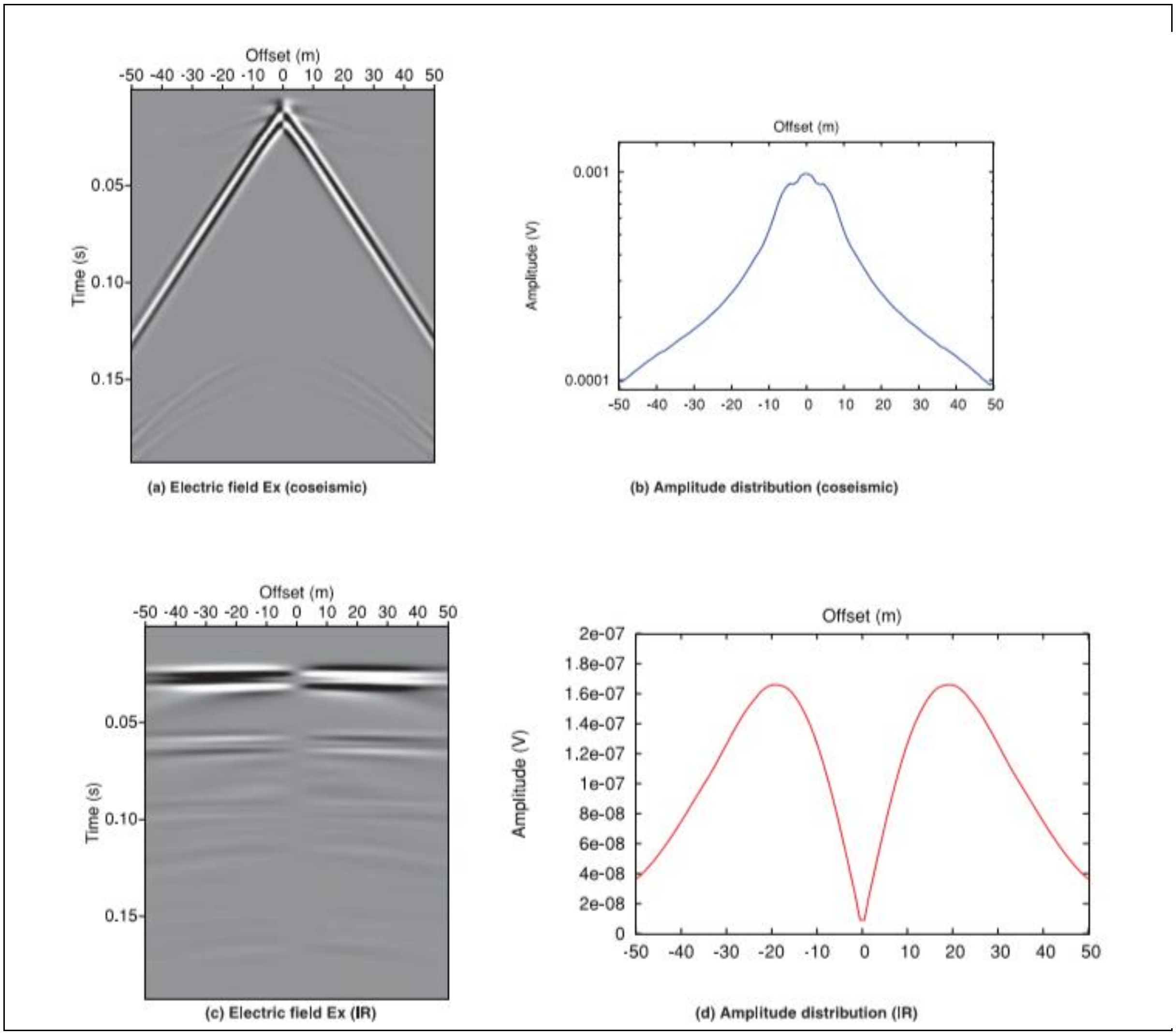


Figure 5: (a) Synthetic recording of the horizontal inline electric field reproduced from Figure 4.(b) Amplitude distribution of recording (a): the maximum absolute amplitude for each trace has been picked and plotted against offset. (c) Synthetic recording of the horizontal inline electric field for which the interface response was artificially highlighted. (d) Amplitude distribution of recording (c): the maximum absolute amplitude for each trace has been picked and plotted against offset.

Recently, several authors have proposed an alternate formulation of the seismoelectric theory, relying on the quasi-static limit of the electromagnetic fields. For instance, Revil and Jardani (2010) have applied this approach to model the seismoelectric response in heavy oil reservoirs. As highlighted by Grobbe and Slob (2016), this approach allows to simplify the system and might help develop the seismoelectric technique toward inversion. We will not further address this approach but direct the interested reader to the handbook of Revil *et al.* (2015).

Pride's theory relies on a number of assumptions, among which:

- The fluid is an ideal electrolyte, i.e. ions move independently from one another. This hypothesis is usually verified for low salinities (<1 mol/L).
- The medium is isotropic.
- The medium is fully saturated.
- Diffusion effects are negligible.
- The wavelength is much larger than the grain size.

In order to describe the conversion mechanisms between seismic and electromagnetic waves, Pride combined Maxwell's equations with Biot's equations for poroelasticity describing the propagation of seismic waves throughout a saturated porous medium. These two subsets of equations are linked through the two following coupling equations:

$$\mathbf{J} = \sigma(\omega)\mathbf{E} + L_{12}(\omega)\left(-\nabla p + \omega^2 \rho_f \mathbf{u_s}\right)$$ *Equation 2*

$$-i\omega\mathbf{w} = L_{21}(\omega)\mathbf{E} + \frac{k(\omega)}{\eta}\left(-\nabla p + \omega^2 \rho_f \mathbf{u_s}\right)$$ *Equation 3*

J is the electrical current density [$A.m^{-2}$], expressed as the sum between conduction and electro-filtration currents. The first term on the right-hand side of Equation 2 corresponds to ohmic currents: σ is the conductivity [S/m] and E the electric field [V/m]. The second term on the right-hand side of Equation 2 is related to the electro-filtration currents. These may be induced by fluid pressure p [Pa] or by the acceleration of the solid matrix $\omega^2\rho_f u_s$[$m.s^{-2}$], where ω is the angular frequency [$rad.s^{-1}$], ρ_f is the fluid volumic mass [$kg.m^{-3}$] and u_s is the displacement of the solid phase. L_{12} is the seismoelectric coupling coefficient; note that when L_{12}=0, Equation 2 falls back to Ohm's law.

w[m] is the relative displacement between fluid and solid phases; its derivative –iωw [m/s] corresponds to Darcy's velocity. η [Pa.s] is the viscosity and k is the frequency-dependent dynamic permeability [m^2]. L_{21} is the electro-seismic coupling coefficient.

Pride derived that L_{12}=L_{21}, i.e. the seismoelectric coupling equals the electro-seismic coupling. This is called the Onsager reciprocity and was already verified for electrokinetic and electro-osmotic phenomena. One can therefore write: L_{12}=L_{21}=L.
This coupling coefficient L(ω) can be written as the product of a static tem L_0 with a frequency-dependent term.

Field equipment and processing

In this section, we present a dataset acquired in a sedimentary environment by Garambois & Dietrich (2001), as well as the key processing steps applied to this data, as described in Warden *et al.* (2012).

Source and receivers key features

The minimal field equipment for a seismoelectric experiment consists of:

- A seismic source: a plate hit by a sledgehammer, a weight drop or even a vibrating source, as detailed in the next section.
- Sensors: pairs of electrodes forming a dipole over which a voltage can be measured.
- Cables conveying the signal back to the recording system.
- The recording system itself.

- Preamplifiers are sometimes used to buffer the signal before it is recorded.

In order for the digitizer not to affect the current travelling within the subsurface between the electrode pair, its resistance must be much higher than that of the ground. However, the ground contact resistance may reach several tens of kΩ which may exceed the device's input impedance, which must therefore be increased through the use of pre-amplifiers. Pre-amplifiers also allow to multiply the weak (several mV to several µV) seismoelectric signals before those undergo a stacking procedure. On the other hand, they ensure a high input impedance.

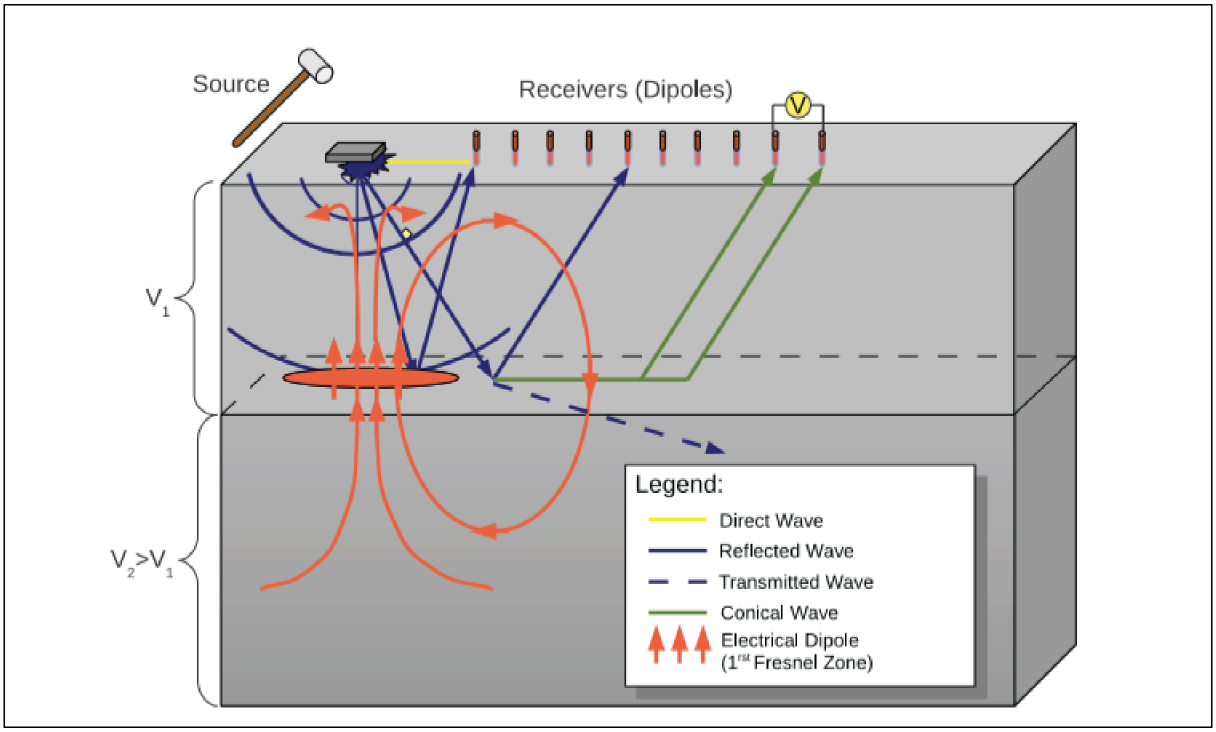


Figure 6: Typical seismoelectric layout. From Warden et al., 2012.

Acquisition

In Figure 7 we present seismoelectric data acquired in a sedimentary environment by Garambois & Dietrich (2001).
These data were collected in a river catchment area by the Fier river near the Annecy area. The stratigraphy of the site is marked by alluvial deposits with alternating layers of gravel, sand, and clay from the surface down to 150m depth. At the time of the experiment, the water table was located at a depth of about 1.5m. The source used here was a 200g dynamite charge buried at a depth of 1m. The data were recorded by 12 unevenly spaced dipoles, deployed on either side of the shot point, between offsets x=±5m and x=±30m.

Data processing: Anthropic noise reduction

Seismoelectric data are often contaminated with anthropic noise, mainly 50 or 60Hz power-line noise, depending on the survey area.

Several techniques have been proposed to reduce this power-line noise.

The **block subtraction technique** (Butler & Russell, 2003) implies recording for a few seconds while the seismic source is not active, typically just before activating the source. This "passive" recording, assumed to consist of pure harmonic noise, is used to estimate the ambient noise: it can then be duplicated over the entire duration of the actual seismoelectric recording before being subtracted from the data.

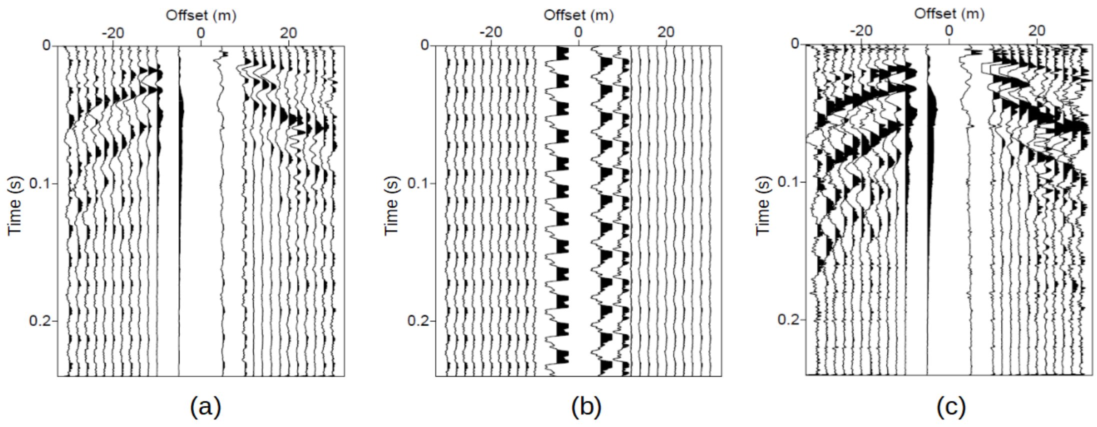


Figure 7: Seismoelectric data acquired by the Fier river. From Garambois & Dietrich, 2001. Results of the sinusoidal subtraction technique: (a) raw data. This recording is dominated by coseismic Rayleigh surface waves.; (b) power line noise estimate; (c) processed data. Power line noise was estimated up to the order n=15.

The drawback with this simple approach is that any non-harmonic noise component present over the initial recording interval will be subtracted from the data and will thus contaminate the processed recording, where it will appear with opposite polarity.

Also, extra care must be taken when duplicating the initial noise interval over the entire duration of the seismoelectric recording. Ideally, the noise block should be shifted by a multiple of $1/f_0$, where f_0 is the fundamental powerline noise frequency: this step may require resampling the data.

For the example given in Figure 7, we did not have access to data acquired before the source was activated: we therefore assumed the last 40ms of the recording consisted of pure harmonic noise. The noise estimate is represented in Figure 7(b). The processed data appear in Figure 7(c).

Minimizing the power-line noise magnifies some of the seismoelectric arrivals at late time and large offsets.

The **sinusoidal subtraction technique** consists in finding the amplitude and the phase of the *n*-th noise harmonic contaminating the seismoelectric recording through a least-square approach. The harmonic noise *p(t)* can be written as a sum of cosine and sine functions such as:

$$p(t) = \sum_{k=1}^{\infty} a_k \cos(kw_0 t) + b_k \sin(kw_0 t)$$

Equation 4

Where t is the time, ω_0 is the fundamental angular frequency and k is the order of the harmonic.

For each order k, the amplitude and phase of the *k*-th harmonic can be deduced from the coefficients a_k and b_k. These coefficients are unknown, but can be estimated by minimizing the difference between the data and the sinusoid of frequency nf_0 in a least-square sense.

Data processing: Interface response extraction

The previous section describes how to reduce the power-line noise often contaminating seismoelectric data.
The processed seismoelectric recording contains signals of both types: type I (coseismic waves) and type II (interface response), as described in the section **Back to the Theory**.
The coseismic signals provide information about the medium in the vicinity of the receivers, that is, in the very near surface. The interface response, on the other hand, provides information about the subsurface at depth.

Unfortunately, the interface response may be hard to recover, because its amplitude is generally much weaker than that of the coseismic response. Furthermore, in the case of surface geometries, the interface response usually shows up at about half the two-way travel time needed by the associated coseismic wave to reach the receivers: the total travel time of the interface response corresponds to the time needed for the direct seismic wave to reach the interface, plus the time needed by the converted electromagnetic wave to travel back to the surface, usually much smaller. Therefore, the weak interface response is often "masked" by the stronger coseismic signals, as seen in Figure 8 in the case of synthetic data.

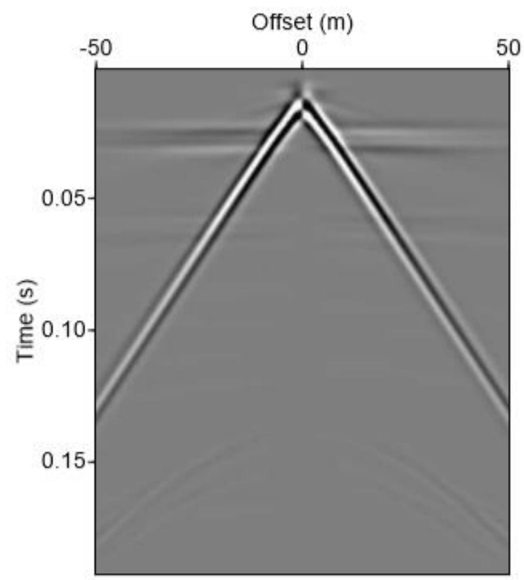


Figure 8: Synthetic seismoelectric recording modeled for a two-layers tabular medium consisting of a sand layer on top of a sandstone half-space. The horizontal event seen at about t=0.02s corresponds to the interface response. From Warden et al., 2012.

The interface response can be extracted through digital filtering techniques. Most of these filtering strategies are based on the interface response's very high velocity, which translates into the fact that the interface response appears as a horizontal event on the seismoelectric recording.

Frequency-wavenumber (f − k) domain filtering is one of these dip-based techniques: this method commonly applied to seismic data to separate events based on their slope. For instance, it is used to filter out the ground roll from seismic data. Several authors (Haines et al. 2007; Strahser 2007; Warden et al., 2012) have applied this strategy to IR extraction.

Let us consider the same seismoelectric dataset that was processed for power-line noise in the previous section: this recording is represented in Figure 9 a.

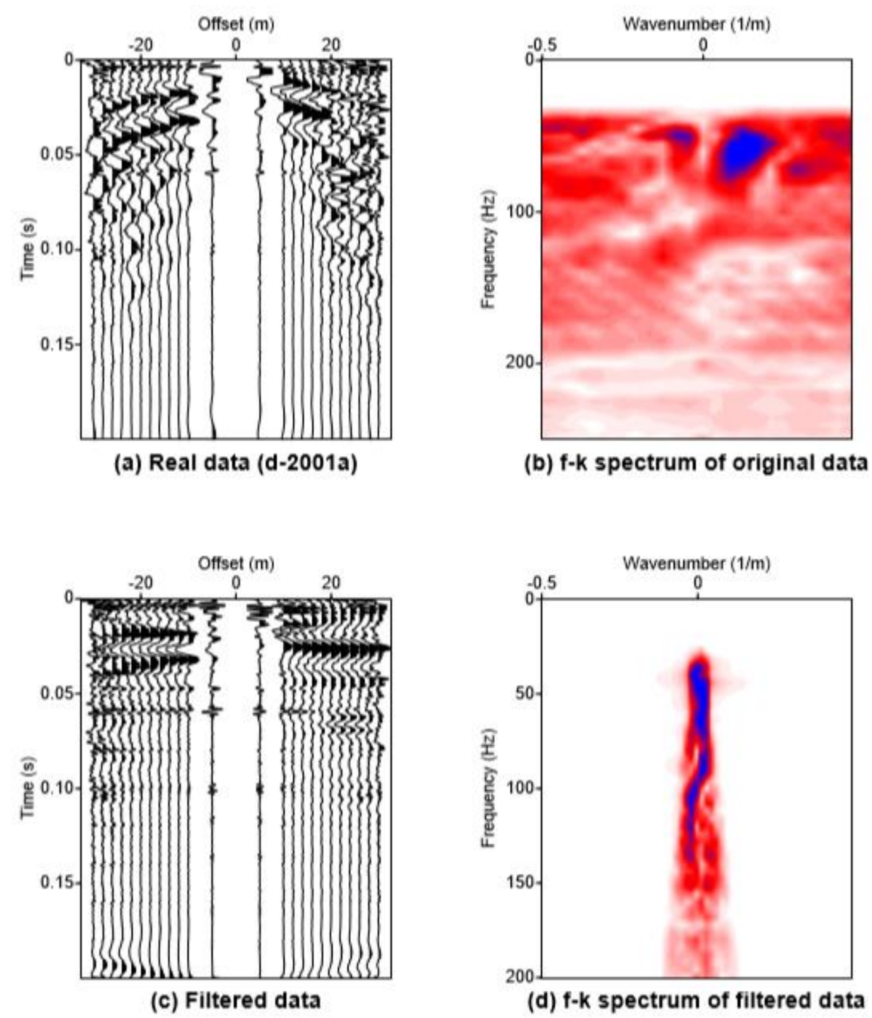


Figure 9: (a) Original seismoelectric dataset after power-line noise removal, represented in the time-distance (t-x) domain. (b) Dataset from a) represented in the frequency wavenumber domain. (c) Data in t-x domain after IR extraction. (d) Data in f-k domain after IR extraction using a "pie-slice" filter selecting high-velocity events.

f-k filtering consists in applying a spatial and temporal Fourier transform to the original data: the data in f-k domain are represented in Figure 9 b. In this representation, events are plotted along lines whose slopes are given by their apparent velocities v=df/dk. Thus, zero-slowness arrivals such as type II converted waves align along the vertical wavenumber axis k=0 and can be recovered using a "pie-slice" filter; see for instance Figure 9 d. The data are then Fourier transformed back to the conventional time-space (t − x) domain, as shown in Figure 9 c. Interface responses can be clearly seen on selected time windows after f-k filtering (figure 10).

f-k filtering is easy to apply but suffers a number of limitations: it requires a good spatial resolution, i.e. a large number of traces, and it does not filter out the flat portions of the type I coseismic signal, such as reflection hyperbolas near the shot-point. Last but not least, it may alter the amplitudes of the extracted IR.

Other filtering techniques (Warden *et al.*, 2012) have been devised to circumvent these problems: some of them are dip-based, such as Radon domain filtering, while others rely upon filtering in the curvelet domain. Curvelets are a type of wavelets that allow to optimally describe seismic or seismoelectric wave-fronts. After decomposing both the seismic and seismoelectric recordings using this multi-scale and multi-direction transform, it is then possible to apply a mask in the curvelet domain taking advantage of the correlation between seismic acceleration and coseismic data.

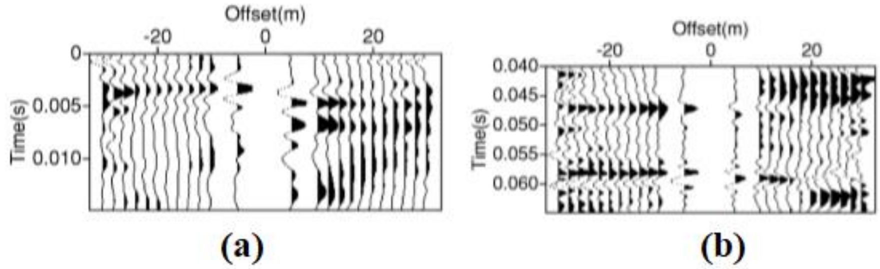


Figure 10: Close-up view on Figure 9 c. These two figures show some of the interface responses extracted through f-k filtering. Note the opposite polarities on either side of the shot point at offset x=0.

Last but not least, some authors (e.g. Haines, 2004) have suggested using fan-shaped acquisition geometries to achieve "natural" separation between coseismic and interfacial signals. Setting source and receivers on either side of the interface of interest allows the interface response to arrive before the associated coseismic waves.

Survey example

We present here an experiment designed to overcome the low signal-to-noise ratio (SNR) associated with seismoelectric conversions. The use of limited energy sources such as a weight drop or a sledgehammer hitting a plate usually produces weak signals and requires a large amount of stacking to achieve an acceptable SNR, even at shallow depths of several meters. In this experiment, resorting to a hydraulic vibrator helped overcome this difficulty.

This work was a joint effort between WesternGeco, Schlumberger Water Services and the Curtin University; it was presented at the SEG 2012 annual meeting.

Experiment layout

This "brute-strength" experiment was conducted at an aquifer storage and recovery (ASR) site in the Abu Dhabi Emirate. ASR consists in injecting water into an aquifer in order to produce it later on: a wide variety of seismic and non-seismic methods can be used to help delineate the aquifer. In April 2011, a seismic survey was conducted at this ASR site: a seismic receiver line consisting of both one-component (1C) and three-components (3C) geophones was deployed over 1,2km. The source line was 3,6km long, extending 700m West and 1,700m East of the receiver line.

The key idea behind this experiment was to record seismoelectric data simultaneously with seismic data to take advantage of the powerful seismic source used for seismic imaging. Three separate seismoelectric spreads were deployed, each array consisting of 49 electrodes separated by 4m. The first two spreads were deployed at the eastern end of the seismic receiver line, while the third spread was centered on the receiver line.

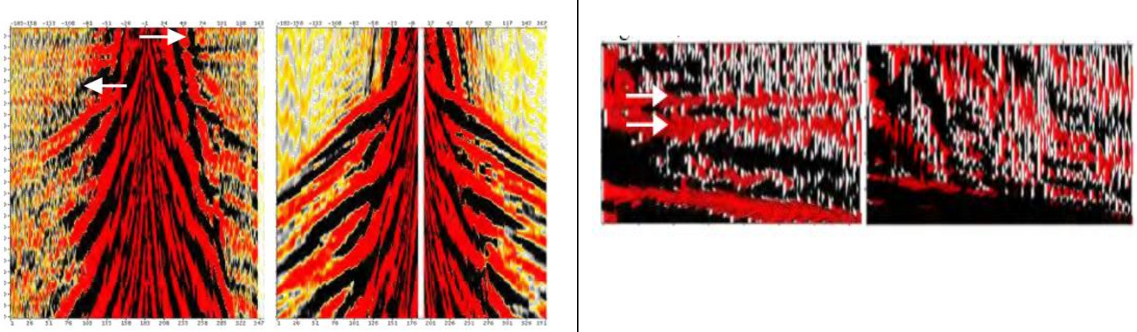
The source used here was an 80,000lb peak force tracked DX-80 Desert Explorer Vibrator.

Data processing

50Hz power-line noise was the main noise source here. It was attenuated using a harmonic subtraction algorithm such as the one described in Section "Data processing: Anthropic noise reduction". Vibroseis correlation was also performed on the data.

Results

Figure 11-a shows the seismoelectric and seismic super-gathers created from 8 records acquired with a 1m source spacing. Both recordings look fairly similar as several seismic events have seismoelectric counterparts, exhibiting similar slope and intercept time values: these are coseismic or type-I waves. But horizontal arrivals can also be seen on the seismoelectric recordings, as highlighted by the white arrows at 32ms and 47ms: these arrivals correspond to waves travelling at velocities several orders of magnitude greater than seismic velocities. Also, they exhibit reversed polarities on either side of the shot point at x=0. Last but not least, they appear at early times, that is, before the reflected coseismic waves reaches the receivers at the surface. These characteristics indicate that these arrivals correspond to the type-II interface response.



a

b

Figure 11: Field example. (a) Seismoelectric (left) and seismic (right) data super-gathers. These gathers were obtainedd by combining 8 records. The trace spacing is 1m, with a maximum offset at 160m. The recording duration is 0.3s. (b) Close-up view on the recordings from Figure 11 a: seismoelectric traces are displayed on the left-hand side, while seismic traces are seen on the right-hand side. The first 100ms are displayed here, for offsets comprised between x=60 and x=100 m. The traces are sorted by offset. The white arrows indicate two interface responses.

Figure 11.b shows a close-up view of the recordings seen in Figure 11.a. The first 100ms for offsets comprised between x=60 and x=100m are displayed here for the seismoelectric recording on the left-hand side. The corresponding seismic recording is displayed to the right. The white arrows indicate two interface responses.

Note that these events of zero slowness have no equivalent in the seismic recording. The recording shown here is a super-gather, which means that it is a combination a several recordings: random electrical noise is cancelled through this stacking process which confirms the flat events highlighted here are indeed interface responses.

Acknowledgements

The author would like to thank Stéphane Garambois and Tim Dean for allowing to reproduce figures from their work.

Suggested readings

A comprehensive review of the seismoelectric method was recently written by Jouniaux & Zyserman (2016). "A review on electro-kinetically induced seismo-electrics, electro-seismics, and seismo-magnetics for Earth sciences". This work compiles several field and lab examples and is a good starting point for the interested reader.

For those who wish to go further, we also suggest reading the handbook "The Seismoelectric Method: Theory and Application" by Revil *et al.* (2015).

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